IDS 701

Unifying Data Science

MIDTERM STUDY GUIDE

Comprehensive Reference — All Lecture Content

Exam: ~15 questions | 1.5 hours | Multiple Choice + Short Answer

1. What Is Data Science & the Problem-Solving Framework

Data science is the study of how to solve problems by answering questions using quantitative methods. Every data science tool can be viewed as a method of answering questions about the world as embodied in your data.

The formal expression of this:

$$\underset{a \in \text{possible answers}}{\text{argmin}} \ \text{Loss Fn}(a)$$

The answer to any data science question is the value that minimizes the loss function over all possible answers.

1.1 The Problem-Solving Process

Step 1: Work with stakeholder to articulate the problem you wish to solve.

Step 2: Determine what question, if answered, would help address the problem. Three types of questions:

Question Type	Description & Use Case
Descriptive Questions	Describe the overall distribution of features in the world. Useful for prioritizing efforts (e.g., where is disease most prevalent?).
(Passive) Prediction Questions	Predict future or unknown outcomes for individuals. Useful for identifying individuals for attention (patients likely to have complications) or automation (rate a job applicant).

Question Type	Description & Use Case
Causal Questions	Predict the likely consequence of taking an action. Useful when you want to intervene and change something.

Step 3: Collect data and implement the chosen method.
Step 4: Evaluate results and translate back to stakeholder's problem.

2. Why Prediction Is Not Enough — Causal Inference

🔑 **Core Insight**
When you take an action, you change the data generating process. Correlations that existed in historical data may no longer apply after the intervention. This is why correlation does not necessarily imply causation.

2.1 The Hotel Price Example

If you regress hotel room prices on occupancy rates using historical data, you'll find higher prices are correlated with higher occupancy. Should you raise prices to increase occupancy?

No. In historical data, prices were high because external demand was already high — something else was driving both. If you unilaterally raise prices, that external demand driver is gone. Causation runs the opposite direction from what correlation suggests.

🔑 **Key Takeaway**
Causal inference is about understanding when correlation does imply causation, and why. Prediction tells us what IS; causal inference tells us what WILL BE if we act.

3. Statistical Decision-Making & Uncertainty

3.1 Frequentist Hypothesis Testing — Problems

The classic approach: reject the null hypothesis H_0 if $p <$ threshold (usually 0.05).

Problem	Explanation
H_0 is a straw man	The null of exactly zero effect is almost never true. We're asking if something is *exactly* zero, which is an unrealistically precise question.
Binary decision	$p = 0.049$ and $p = 0.051$ are treated completely differently despite being nearly identical. As the saying goes: 'Surely God loves the .06 nearly as much as the .05.'
Conflates magnitude and precision	A tiny effect measured very precisely will be 'significant.' A large effect measured imprecisely

Problem	Explanation
	won't be. Statistical significance does NOT mean substantively important.
Sample size dependence	There will always exist a sample size large enough to reject H_0 of zero effect. ALWAYS. This means with enough data, any tiny effect becomes 'significant.'

3.2 Three Statistical Frameworks

Framework	Core Idea
Frequentist (Hypothesis Testing)	θ is a fixed (unknown) constant — not a random variable. Uncertainty comes from sampling variation. Binary: reject or don't reject H_0.
Maximum Likelihood Estimation (MLE)	Still frequentist. Finds the 'most likely' value of θ. Asymptotically gives a sampling distribution P(data | θ). Foundation for confidence intervals.
Bayesian	θ IS a random variable. We estimate the distribution of likely values. Gives P(θ | data) — which is intuitively what we want. Explicitly incorporates prior beliefs.

3.3 P-Values — What They Actually Are

What we WANT	`Pr(H₀ true | Observed Data)`	*Probability the null is true given what we saw*

What p-value IS	`Pr(Observed Data | H₀ true)`	*Prob of seeing data this extreme IF null were true*

These are not the same! How do they relate? Via Bayes' Rule:

Bayes' Rule	`Pr(A|B) = Pr(B|A) ·Pr(A) / Pr(B)`	*Relating conditional probabilities*

$$\Pr(H_0 \text{ true} \mid \text{Data}) = [\Pr(\text{Data} \mid H_0 \text{ true}) \times \Pr(H_0 \text{ true})] / \Pr(\text{Data})$$

The p-value (Pr(Data | H_0 true)) is just one component. To get what we actually want, we need to multiply by the prior probability that H_0 is true, which we do not know from the data alone.

> ⚠️ **Classic Mistake**
> A p-value of 0.03 does NOT mean there is a 3% probability the null is true. It means: if the null WERE true, we'd see data this extreme only 3% of the time. These are very different statements.

3.4 Confidence Intervals

In the MLE framework, assuming your model is correctly specified, the coefficient estimate $\hat{\theta}$ has an asymptotically normal sampling distribution:

MLE Sampling Distribution	$\hat{\theta} \sim N(\theta, \sigma^2)$	*$\hat{\theta}$ is normally distributed around the true θ with variance σ^2*

> ⚠️ **The Frequentist Technicality**
> Technically, a 95% confidence interval does NOT mean 'there is a 95% probability the true value lies in this interval.' Frequentists say: if you repeated this study many times, 95% of the intervals you'd construct would contain the true θ. The interval either contains it or doesn't.

3.5 Bernstein-von Mises Theorem

> 🔑 **Theorem**
> The MLE sampling distribution (Pr(data | θ)) converges to the Bayesian posterior distribution (Pr(θ | data)) as n → ∞, given 'reasonable' (non-dogmatic) priors.

Practical implication: In large samples, it is defensible to treat confidence intervals as if they represent the distribution of the likely true value of the coefficient. In small samples, you must be careful.

Convergence is faster when priors are weak (before the analysis, you'd assign similar likelihood to all possible values of θ). Stronger priors → slower convergence. Eventually, enough data overwhelms any prior.

Caveats to this Practical Use of CIs	Why it Matters
Assumes MLE is properly specified	If your model has the wrong functional form or omits key variables, all bets are off.
Not true in small samples!	The convergence hasn't happened yet in small samples.
Ignores model uncertainty	There's uncertainty about which model is correct, not just about the parameters within a model.

4. The Potential Outcomes Framework (Rubin Causal Model)

4.1 The Fundamental Problem of Causal Inference

For each individual i, we define two potential outcomes:

Notation	Meaning
Y_i^1	Potential outcome for unit i IF treated (D = 1)

Notation	Meaning
Y_i^0	Potential outcome for unit i IF NOT treated (D = 0)
D_i	Treatment indicator: 1 if treated, 0 if control
Individual Treatment Effect	$Y_i^1 - Y_i^0$ (we can never observe both for the same person!)

⚠️ **The Fundamental Problem**
We can never observe both Y_i^1 and Y_i^0 for the same individual at the same time. A person either gets the treatment or doesn't. The unobserved outcome is called the counterfactual.

4.2 Treatment Effects — What We're Estimating

ATE	$E(Y^1) - E(Y^0)$	*Average Treatment Effect: avg effect for ALL units in population*
ATT	$E(Y^1 \mid D=1) - E(Y^0 \mid D=1)$	*Average Treatment Effect on the TREATED*
What We Measure	$E(Y^1 \mid D=1) - E(Y^0 \mid D=0)$	*Naïve comparison (treated outcomes vs control outcomes)*

🔑 **The Big Picture**
Our goal is ATE. We can measure $E(Y^1|D=1) - E(Y^0|D=0)$. If NO BASELINE DIFFERENCES: measured value = ATT. If ALSO NO DIFFERENTIAL TREATMENT EFFECTS: measured value = ATT = ATE. 🎉

Why do we care about ATE? Because when we take a policy action (authorize a drug, change a price, change a webpage), we do it for everyone — the average effect will be the ATE.

4.3 Condition 1 — No Baseline Differences

No Baseline Differences	$E(Y^0 \mid D=1) - E(Y^0 \mid D=0) = 0$	*In a no-treatment world, outcomes would be equal across groups*

This says: in a world where no one got the treatment, the average outcome would be the same for those we observed getting treatment and those we observed not getting it. This does NOT require:

- No differences between the D=0 and D=1 groups on any observable characteristics.
- No differences in Y across individuals.

It requires only that the groups are balanced on average potential outcome Y^0.

4.4 Condition 2 — No Differential Treatment Effects

No Differential Tx Effects	$E(Y^1\|D=1)-E(Y^0\|D=1) = E(Y^1\|D=0)-E(Y^0\|D=0)$	*Treatment effect is the same (on avg) for both groups*

If this is NOT met but No Baseline Differences IS met, you still get a valid estimate — but it's the ATT (Average Treatment Effect on the Treated), not the full ATE. ATT answers: what is the average effect for the people who actually received the treatment?

4.5 The Conditional Independence Assumption (CIA)

CIA	$(Y^1, Y^0) \perp\!\!\perp D \mid X$	*Potential outcomes are independent of treatment, CONDITIONAL on X*

This is the statistician's version. It's slightly stronger than 'no baseline differences' — it implies both no baseline differences AND no differential treatment effects, all conditional on covariates X. This is what matching and regression try to satisfy.

5. SUTVA — Stable Unit Treatment Value Assumption

5.1 What SUTVA Requires

SUTVA is a critical assumption lurking in all potential outcomes math. It has two parts:

SUTVA Component	Meaning
Treatment Homogeneity	The treatment is the same thing for every unit. There is only one version of 'treatment.'
No Interference	The treatment status of one unit does NOT affect the potential outcomes of another unit.

> ⚠️ **When SUTVA Is Violated**
> If treated and control units interact with each other, the outcomes of control units are contaminated by the treatment. This can make treatment effects appear smaller OR larger than they really are.

5.2 Classic SUTVA Violation Examples

Vaccine Study at Duke

Problem: Half of Duke students get the vaccine, half get placebo. But now vaccinated students are less likely to spread COVID to their unvaccinated peers. The control group's outcomes are improved by the treatment group, making the vaccine look less effective than it is.

Fix: Vaccinate or not vaccinate entire dorms (cluster randomization), or use a much smaller treatment proportion, or randomize across geographically separate communities.

Uber Driver Pay Experiment

Problem: 10% of Durham drivers get higher pay → more treatment drivers go online → harder for control drivers to get rides. Also: the treatment drivers have less competition than they'd have if ALL drivers got the higher pay.

Two effects: (1) control drivers are harmed artificially → effect appears larger than true ATE. (2) treatment drivers face less competition than full rollout → ATE estimate from experiment may be inflated.

Fix: Assign entire cities to treatment or control. Or randomize across different time periods.

Instagram Feed Algorithm

Problem: 10% of users get only 'happy content.' But users interact with each other. A control user might get a post from a treated user who's posting more happily. Their behavior changes due to treated users' behavior, contaminating the control condition.

6. Experiments & Internal Validity

6.1 Why Experiments Work for Causal Inference

Randomization, combined with the Law of Large Numbers, ensures the causal assumptions hold:

Assumption	How Experiments Satisfy It
No Baseline Differences	Random assignment ensures treated and control groups are (probabilistically) identical on all characteristics — observed AND unobserved. Guaranteed by design.
No Differential Treatment Effects	Same randomization argument — groups are equivalent, so treatment effects should be similar.
SUTVA / No Interference	NOT automatically satisfied by randomization. Requires careful choice of randomization unit and domain expertise.

6.2 Internal Validity Checks for Experiments

After running an experiment, verify internal validity with these checks:

Check	What It Tests
1. Balance Tests	Compare pre-treatment covariates across treatment and control arms. They should be statistically indistinguishable. Imbalance suggests the randomization failed.
2. Sample Ratio Check	Verify that the number of units in treatment vs. control matches what the randomization probability would predict. If 50/50 split was intended and you got 60/40, something went wrong.

Check	What It Tests
3. A/A Test	Run an experiment where BOTH groups get the control condition. Any measured 'effect' is spurious and reveals problems in the measurement infrastructure itself.

7. External Validity

🔑 **Definition**
External validity is the extent to which the effect measured in a study is a good estimate for the effect in a DIFFERENT context. It is NOT singular — a study can have high external validity for one context and low external validity for another.

External validity is always defined relative to the context you wish to generalize to.

7.1 Simple Sources of External Validity Failure

Source	Explanation & Examples
Sample Composition	Who was in the study? Medical studies historically over-represent white men → cardiac drug doses often too high for women, some drugs interact poorly with genes common in Asian/Pacific Islanders, MS is driven by different mutations in Black vs. European descendants. Psychology studies mostly use WEIRD subjects (Western, Educated, Industrialized, Rich, Democratic countries).
Treatment Differences	Is the treatment in the new context the same? A $3,000 discount mailer result doesn't necessarily generalize to a $1,000 discount, a used car dealer, or hand-delivered coupons.
Broader Context / Artificiality	More controlled = more internal validity but sometimes less external validity. Hawthorne Effect: people behave differently when they know they're being observed. Originally documented in factory worker studies.

7.2 Short-Term vs. Long-Term: Time-Shifting & Competition

Promotions and temporary treatments affect consumers in at least two ways:

- Win customers from competitors: these are REAL gains (though competitors may eventually respond).
- Time-shift purchases: customers who would have bought next month buy this month. These are ILLUSORY gains — they disappear in the long run.

Short-run A/B test may look positive because of time-shifting, but rolling out permanently yields no long-run benefit. Primacy and novelty effects are other reasons short-term results may not hold long-term. The relative magnitude of each effect depends on product type and market structure.

Example — Bing vs. Google: If Bing degrades search quality, users search more per query (seeing more ads). Bing can afford this because switching costs are low enough that users tolerate it. Google cannot, because users will switch to a competitor (they have more to lose). Same experiment, different external validity to long-run deployment.

7.3 General Equilibrium Effects

Even if a study perfectly captures the short-run causal effect, rolling out a treatment at scale can change the world in ways the study didn't capture.

Type of GE Effect	Example
Political Responses (Rajasthan)	A program to improve nurse attendance worked in the experiment. But when rolled out, local politicians and administrators had incentives to undermine it, so the effect diminished.
Adversarial Users	People change their behavior specifically to game the system. SEO is the quintessential example — website operators optimize for the algorithm, reducing the quality of rankings.
Financial Fraud Adaptation	Fraud detection systems train on historical fraud. Fraudsters adapt: e.g., keeping purchases just under $10,000 threshold for reporting, or stealing tiny amounts from many accounts.
Content Moderation Adaptation	When platforms ban certain words ('unalive' instead of 'suicide,' 'seggs' instead of 'sex'), users create new code words. The censorship algorithm is outpaced.
Surge Pricing Gaming	Rideshare drivers coordinate to sign off simultaneously, creating artificial 'shortages' that trigger surge pricing, then come back on.

7.4 Complete External Validity Recap

- External validity is NOT singular — always relative to a specific generalization context.
- Simple considerations: sample composition, treatment substance, broader context/artificiality.
- Short vs. long term: primacy/novelty effects, time-shifting vs. real competition gains.
- General equilibrium effects: political responses and adversarial users.

8. Observational Data, Regression & Omitted Variable Bias

8.1 The Challenge of Observational Data

In experiments, randomization ensures no baseline differences by the Law of Large Numbers. Without randomization, we must rely on observational data, where treated and control units can differ systematically.

Strategy: If we understand the source of baseline differences, we can condition on covariates X such that:

Conditional No-Baseline-Differences	$E(Y^0 \mid D=1, X) = E(Y^0 \mid D=0, X)$	*Within levels of X, no baseline differences*

This is what regression and matching both try to achieve.

8.2 Omitted Variable Bias (OVB) — The Key Formula

Suppose the TRUE causal model (the 'long' regression) is:

$$Y_i = \alpha L + \beta L \cdot D_i + \gamma \cdot A_i + \varepsilon_i \qquad \text{(Long/True Model)}$$

But you estimate the 'short' regression (omitting A):

$$Y_i = \alpha S + \beta S \cdot D_i + \varepsilon_i \qquad \text{(Short/Estimated Model)}$$

OVB Formula	$\beta L - \beta S = \pi_1 \times \gamma$	*Bias = (relationship between A and D) × (effect of A on Y)*

Where π_1 comes from the auxiliary regression: $A_i = \pi_0 + \pi_1 \cdot D_i + \mu_i$

Variable	Meaning
π_1	How much A and D are correlated (how much does A change when D changes?)
γ	The effect of the omitted variable A on outcome Y
$\beta L - \beta S$	The omitted variable bias — how much your estimate is wrong by

> ⚠️ **When Does OVB Occur?**
> OVB requires BOTH conditions: (1) the omitted variable A is correlated with the treatment D ($\pi_1 \neq 0$), AND (2) A has an effect on Y ($\gamma \neq 0$). If either is zero, there's no bias. Think of it as: there must be a reason why people with different underlying Y^0 sort into D=0 vs D=1.

OVB is just a formalization of the 'baseline differences' problem: $E(Y^0|D=1) \neq E(Y^0|D=0)$. The omitted variable A is what creates those baseline differences.

Direction of bias: if π_1 and γ have the same sign → positive bias (βS overestimates true effect). If opposite signs → negative bias (βS underestimates).

9. Functional Forms & Matching

9.1 Functional Form Assumptions in Regression

Every regression specification makes implicit assumptions about how variables relate to outcomes. Linear regression assumes additivity and linearity. Consider:

$$\text{delinquency}_i = \beta_0 + \beta_1 \cdot \text{credit_score}_i + \beta_2 \cdot \text{income}_i + \beta_3 \cdot \text{interest_rate}_i + \varepsilon_i$$

This model assumes:

- A $10K income increase reduces delinquency risk by the same amount regardless of whether income goes from $20K to $30K or from $100K to $110K (linear income effect).
- The effect of interest rates is the same for every borrower regardless of their income (no interaction).

Using log transformation:

$$\text{delinquency}_i = \beta_0 + \beta_1 \cdot \text{credit_score}_i + \beta_2 \cdot \ln(\text{income}_i) + \beta_3 \cdot \text{interest_rate}_i + \varepsilon_i$$

Now a 1% income increase always changes delinquency by the same absolute amount — a $10K increase from $20K is much larger (50%) than from $100K (10%), so it has a bigger effect. This is often more realistic for income.

> 🔑 **Rule of Thumb**
> Choosing the wrong functional form is itself a type of model misspecification that can bias your estimates, even if you have good data and no omitted variables.

9.2 Matching — Promise and Limitations

Matching tries to create a control group that looks like the treatment group, without assuming a specific functional form. In principle, it 'lets the data speak' about how features relate to outcomes.

Matching Type	How It Works
Coarse Exact Matching (CEM)	Bin each covariate into discrete buckets. Match treated and control units that fall in the same bucket. Exact within bins.
Nearest Neighbor Matching	For each treated unit, find the control unit with the most similar covariate vector (by some distance metric).
Propensity Score Matching	Collapse all covariates into a single score (propensity score) and match on that scalar.

But matching on multiple covariates still requires you to define a similarity measure — which implicitly assigns relative importance weights to different features. That is itself a functional form choice.

10. Curse of Dimensionality & Propensity Scores

10.1 The Curse of Dimensionality

With coarse exact matching (CEM), as you add more covariates, the number of possible 'cells' grows exponentially, but your sample size stays fixed. Most cells become empty.

# Covariates (each with 4 values)	# Possible Cells	Avg T per Cell (n=100)
2	$4^2 = 16$	100/16 = 6.25
3	$4^3 = 64$	100/64 = 1.56
4	$4^4 = 256$	100/256 = 0.39
5	$4^5 = 1{,}024$	100/1,024 ≈ 0.10

With just 5 binary covariates, most cells are empty. You can't find matches. This is the curse of dimensionality: adding more dimensions makes points exponentially farther apart in covariate space. This applies even with continuous variables — as dimensions grow, nearly all data points become equidistant from each other, making 'nearest neighbor' meaningless.

10.2 Propensity Scores

The propensity score collapses the high-dimensional covariate vector into a single scalar, avoiding the curse of dimensionality.

Propensity Score	$p(X) \equiv Pr(D_i = 1 \mid X_i)$	*Probability of receiving treatment given observed covariates*

🔑 **Propensity Score Theorem**
If $(Y^1, Y^0) \perp\perp D \mid X$ (CIA holds for the full covariate vector X), then $(Y^1, Y^0) \perp\perp D \mid p(X)$ as well. You can match on the scalar p(X) instead of the full vector X.

Why Does This Work?

Conditioning on p(X) = Pr(D=1|X) breaks the correlation between X and D. Among units with the same propensity score, treatment assignment is as good as random with respect to X. There's no Y^0 in the propensity score equation — it's purely about what predicts treatment selection.

Propensity Score Matching	Description
Strength 1	Matching on a single scalar is computationally easy.
Strength 2	Asymptotically breaks the correlation between X_i and D_i.
Weakness 1	You're not directly making pairs comparable on Y_i^0, Y_i^1 — so the matched sample may be unbalanced and inefficient.
Weakness 2	Proofs assume you know the true Pr(D=1|X). In practice, you must estimate it, introducing additional model error.

🔑 **Instructor's Recommendation**
If you have enough data to match on covariates directly, do that. Use propensity score matching only if (a) you have a strong understanding of the true treatment assignment model and (b) you aren't dropping too many observations.

11. Matching — Distance & Standardization

11.1 The Problem of Scale in Distance Metrics

When calculating distance between observations for matching (e.g., Euclidean distance), variables on larger scales dominate the distance calculation.

Example: If x_1 has standard deviation $\sigma_1 = 3$ and x_2 has $\sigma_2 = 1$, differences in x_1 are counted as if they're 3x more important than differences in x_2 — even if they're equally important for the outcome. An outlier in x_1 can shift which unit gets matched with which.

Euclidean Distance	$d = \sqrt{[(x_{1i} - x_{1\square})^2 + (x_{2i} - x_{2\square})^2]}$	*Raw Euclidean — sensitive to scale*

Solution: Standardize each variable before computing distances, or use Mahalanobis distance which automatically accounts for the covariance structure.

Key insight: The goal of matching is to find units with the same potential outcomes (Y^0, Y^1), not just units that are close in raw covariate space. An outlier in your data can shift which units get matched even if that outlier will be discarded from the matched sample — which is wrong.

12. Complete Formula Quick Reference

All Critical Formulas at a Glance

Data Science Optimization	`argmin_{a ∈ answers} Loss Fn(a)`	*Every DS question minimizes a loss function*

P-value Definition	`p-value = Pr(Data | H₀ true)`	*NOT Pr(H_0 true | Data)*
Bayes' Rule	$Pr(A\|B) = Pr(B\|A) \cdot Pr(A) / Pr(B)$	*Connects p-value to what we want*
MLE Sampling Dist.	$\hat{\theta} \sim N(\theta, \sigma^2)$	*Asymptotically normal around true value*
ATE	$E(Y^1) - E(Y^0)$	*Average Treatment Effect (population)*
ATT	$E(Y^1\|D=1) - E(Y^0\|D=1)$	*Avg Treatment Effect on the Treated*
Measurable Naive ATE	$E(Y^1\|D=1) - E(Y^0\|D=0)$	*What we actually measure from data*
No Baseline Diff.	$E(Y^0\|D=1) = E(Y^0\|D=0)$	*Key assumption for causal inference*
No Diff. Tx Effects	$E(Y^1-Y^0\|D=1) = E(Y^1-Y^0\|D=0)$	*Same treatment effect for both groups*
CIA (Conditional Indep.)	$(Y^1, Y^0) \perp\perp D \mid X$	*Potential outcomes indep. of treatment given X*
Propensity Score	$p(X) = Pr(D=1 \mid X)$	*Probability of receiving treatment*
OVB Formula	$\beta L - \beta S = \pi_1 \times \gamma$	*Bias = Corr(A,D) × Effect of A on Y*
OVB Auxiliary Reg.	$A_i = \pi_0 + \pi_1 \cdot D_i + \mu_i$	*Determines π_1: how correlated is A with D?*

13. Exam Prep — Key Concepts, Common Traps & Practice

13.1 Most Important Conceptual Distinctions

Concept A	Concept B — How They Differ
Internal Validity	External Validity — Internal: did you measure the causal effect correctly within your study? External: does that measured effect generalize?
ATE	ATT — ATE: average effect for entire population. ATT: average effect only for people who actually received treatment. Equal only if No Differential Treatment Effects.
p-value = Pr(Data|H_0)	What we want = Pr(H_0|Data) — p-value and Bayesian posterior are related by Bayes' Rule, NOT equal.
Statistical Significance	Substantive Significance — Stat. sig. just means the effect is distinguishable from zero. It says nothing about whether the effect size matters.
Correlation	Causation — Correlation reflects associations in observational data. Causation requires controlling for confounders (CIA) or randomization.
Frequentist CI	Bayesian Credible Interval — Freq. CI: 95% of such intervals cover true θ. Bayesian: 95% probability θ is in the interval. Bernstein-von Mises: these converge in large n.

13.2 Common Exam Traps

⚠️ Trap 1: Misinterpreting p-values
p = 0.04 does NOT mean 'there is a 4% chance the null is true.' It means 'if the null were true, we'd see data this extreme only 4% of the time.' To get Pr(H_0|Data), you need a prior via Bayes' Rule.

⚠️ Trap 2: Equating Statistical and Substantive Significance
With a large enough sample, ANY effect (no matter how tiny) becomes statistically significant. Always ask: is the effect SIZE meaningful, not just detectable?

⚠️ Trap 3: Assuming Experiments Have External Validity
Experiments guarantee internal validity (within the study). They do NOT guarantee external validity. Results may not generalize to other populations, contexts, time periods, or at scale.

⚠️ Trap 4: OVB Direction
Think carefully about the SIGN of π_1 (correlation between omitted variable and treatment) and γ (effect of omitted variable on Y). Same signs → positive bias. Opposite signs → negative bias.

⚠️ Trap 5: SUTVA Violations in Experiment Design
Randomizing individual units when they interact with each other violates SUTVA. Always ask: can treated units affect control units? Solution: cluster or geographic randomization.

⚠️ **Trap 6: Propensity Score Is Not Magic**
Propensity score matching requires you to correctly specify the model for Pr(D=1|X). If you miss important predictors of treatment, the propensity score doesn't fully balance covariates.

13.3 Short-Answer Strategy

For questions about whether a study's results are valid or generalizable, structure your answer around:

- What assumption is at risk? (No baseline differences, SUTVA, external validity...)
- What is the specific mechanism of the violation? (Why exactly does it fail?)
- What direction does this push the bias? (Overestimate or underestimate the true effect?)
- How could the design be fixed? (Cluster randomization, better sample, longer time horizon, etc.)

13.4 Quick Review Checklist Before the Exam

- Can you define ATE, ATT, and explain when they are equal?
- Can you identify when SUTVA is violated and explain the bias direction?
- Do you know the three internal validity checks for experiments?
- Can you identify all 5 categories of external validity threats?
- Can you apply the OVB formula: $\beta L - \beta S = \pi_1 \times \gamma$?
- Do you understand why H_0 of zero effect is a 'straw man'?
- Can you explain the Bernstein-von Mises theorem in plain English?
- Do you know when propensity score matching is appropriate vs. direct covariate matching?
- Can you explain why prediction is insufficient for causal questions (hotel price example)?
- Can you distinguish general equilibrium effects from simple external validity failures?

14. Glossary of Key Terms

Term	Definition
ATE (Average Treatment Effect)	$E(Y^1) - E(Y^0)$; the average effect of treatment across the entire population.
ATT (Avg Tx Effect on Treated)	$E(Y^1\|D=1) - E(Y^0\|D=1)$; the average treatment effect only for those who received treatment.
Adversarial Users	People who change their behavior specifically to game or exploit an algorithm after it is deployed at scale.
A/A Test	An experiment where both groups receive the control. Any 'effect' found is a measurement artifact.
Balance Test	Pre-analysis check that treatment/control groups are statistically equivalent on pre-treatment covariates.
Bayesian Statistics	Statistical framework where parameters are random variables; estimates the distribution Pr(θ|data).

Term	Definition
Bernstein-von Mises Theorem	The MLE sampling distribution converges to the Bayesian posterior as $n \to \infty$ with weak priors.
CIA (Conditional Indep. Assumption)	$(Y^1, Y^0) \perp\perp D \mid X$; potential outcomes are independent of treatment assignment conditional on X.
Confidence Interval	A frequentist interval; in large samples, defensible to treat as distribution of likely true θ values.
Counterfactual	The unobserved potential outcome — what would have happened to a unit under the other treatment condition.
Curse of Dimensionality	As the number of covariates grows, data becomes sparse and distance-based matching fails.
Descriptive Question	A question about the distribution of features in the world; useful for prioritizing efforts.
External Validity	The extent to which a study's results generalize to a different context, population, or time period.
Frequentist Statistics	Statistical framework where θ is a fixed constant; uncertainty comes from sampling variation.
General Equilibrium Effects	System-wide changes that occur when a treatment is deployed at scale, affecting the context of the experiment.
Hawthorne Effect	People behave differently when they know they are being observed, reducing external validity.
Internal Validity	The extent to which a study correctly estimates the causal effect in the study context itself.
MLE (Maximum Likelihood)	Finds the parameter value that maximizes the probability of observing the data; still frequentist.
No Baseline Differences	$E(Y^0 \mid D=1) = E(Y^0 \mid D=0)$; the groups would have equal outcomes absent any treatment.
No Differential Treatment Effects	The treatment effect is the same for both groups on average.
Omitted Variable Bias	Bias in β from omitting a variable A that is correlated with both D and Y; formula: $\beta L - \beta S = \pi_1 \times \gamma$.
p-value	$\Pr(\text{Data} \mid H_0 \text{ true})$; probability of observing data this extreme if the null hypothesis were true.
Potential Outcomes	Y_i^1 and Y_i^0; the outcomes a unit would have under treatment or no treatment, respectively.
Propensity Score	$p(X) = \Pr(D=1 \mid X)$; the probability of receiving treatment given observed covariates.
SUTVA	Stable Unit Treatment Value Assumption; treatment is homogeneous and units don't interfere with each other.

Term	Definition
WEIRD Subjects	Western, Educated, Industrialized, Rich, Democratic — typical psychology study sample, limiting generalizability.

— Good luck on your exam! —

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